CUSTOMER CHURN ANALYSIS AND PREDICTION

USING MICROSOFT POWER BI

Project Report

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Organization:

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Tools Used:

Microsoft Power BI, Power Query, DAX

Dataset:

Telco Customer Churn Dataset

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1. EXECUTIVE SUMMARY

Customer churn is a critical business challenge that directly impacts revenue and customer retention. This project focuses on analyzing customer churn data using Microsoft Power BI to identify factors influencing customer attrition and generate actionable business insights.

The project involved data acquisition, data transformation, KPI development, demographic analysis, tenure analysis, and churn analysis. An interactive dashboard was developed to visualize customer behavior and highlight key churn drivers. The insights obtained from this analysis can help organizations improve customer retention strategies and support data-driven decision-making.

2. PROJECT OBJECTIVES

The objectives of this project were:

- Import and connect customer churn data into Power BI.
- Perform data cleaning and transformation.
- Calculate customer churn metrics using DAX.
- Analyze customer demographics.
- Visualize customer tenure distribution.
- Identify factors contributing to customer churn.
- Develop an interactive dashboard.
- Generate business insights and recommendations.

3. DATASET OVERVIEW

The project utilized a telecommunications customer churn dataset containing customer demographic information, subscription details, service usage, billing information, and churn status.

Dataset Categories

Demographic Information

- Gender
- Senior Citizen
- Partner
- Dependents

Service Information

- Internet Service
- Online Security
- Online Backup
- Device Protection

- Tech Support
- Streaming TV
- Streaming Movies

Subscription Information
- Contract Type
- Tenure

Billing Information
- Monthly Charges
- Total Charges
- Payment Method

Target Variable
- Churn

The dataset contains 7,043 customer records and provides sufficient information for customer churn analysis.

4. METHODOLOGY

A structured analytics approach was followed throughout the project.

Phase 1: Data Acquisition
Importing the dataset into Power BI.

Phase 2: Data Preparation
Cleaning, transforming, and validating data.

Phase 3: KPI Development
Creating business metrics using DAX.

Phase 4: Data Analysis
Analyzing customer demographics, tenure, and churn factors.

Phase 5: Dashboard Development
Building an interactive dashboard using Power BI visualizations.

Phase 6: Business Insights
Interpreting results and generating recommendations.

5. DATA ACQUISITION

The customer churn dataset was imported into Microsoft Power BI Desktop from a CSV file. After importing the data, the dataset structure, column names, and data quality were reviewed to ensure successful data loading and readiness for analysis.

The imported dataset served as the foundation for all subsequent transformation, analysis, and dashboard development activities.

6. DATA PREPARATION

Data preparation was performed using Power Query to improve data quality and ensure accurate reporting.

<u>Activities Performed</u>

- Reviewed dataset structure and column information.
- Removed blank and incomplete records.
- Verified and corrected data types.
- Handled missing values in the TotalCharges column.
- Created a Churn_Flag column using conditional logic:
    - Yes = 1
    - No = 0
- Created tenure bins for histogram-style analysis.
- Validated the transformed dataset before visualization development.

These transformations ensured reliable KPI calculations and dashboard reporting.

7. KPI Analysis

Three key performance indicators were created using DAX measures.

<u>Total Customers</u>

DAX: *Total Customers = DISTINCTCOUNT(Telco_Customer_Churn_Dataset[customerID])*

Purpose:

Calculates the total number of customers in the dataset.

<u>Churned Customers</u>

DAX: *Churned Customers = SUM(Telco_Customer_Churn_Dataset[Churn Flag])*

Purpose:

Calculates the number of customers who left the company.

Churn Rate
DAX: *Churn Rate = DIVIDE([Churned Customers],[Total Customers],0)*
Purpose:
Calculates the percentage of customers who churned.

KPI Results:

- Total Customers: 7,043
- Churned Customers: 1,869
- Churn Rate: 26.54%

These KPI cards were placed at the top of the dashboard to provide an immediate overview of customer retention performance.

8. CUSTOMER DEMOGRAPHICS ANALYSIS

Customer demographics were analyzed using donut charts to understand the composition of the customer base.

Gender Distribution
A donut chart was used to analyze customer distribution by gender and understand overall customer composition.

Partner Status Analysis
A donut chart was created to compare customers with and without partners, providing insights into customer segmentation.

Dependents Status Analysis
A donut chart was developed to visualize customers with and without dependents and understand family-related customer characteristics.

These visualizations help establish a foundational understanding of the customer population.

9. CUSTOMER TENURE ANALYSIS

Customer tenure represents the length of time customers remain subscribed to company services.

A histogram-style column chart was created using tenure bins to visualize the distribution of customers across different subscription periods.

The analysis helps identify customer retention patterns and highlights customer groups that may require targeted retention initiatives. Customers with shorter tenure periods generally demonstrate a higher risk of churn compared to long-term customers.

10. CHURN ANALYSIS AND DASHBOARD DEVELOPMENT

The dashboard was developed using a single-page layout that combines KPI cards and visualizations to provide a comprehensive view of customer churn.

Dashboard Components

1. KPI Cards

- Total Customers
- Churned Customers
- Churn Rate

2. Customer Demographics

- Gender Distribution
- Partner Status
- Dependents Status

3. Customer Tenure

- Tenure Histogram

4. Churn Analysis

- **Churn by Contract Type**
  A stacked bar chart was used to analyze churn across different contract categories.

- **Churn by Payment Method**
  A stacked bar chart was developed to compare churn behavior across payment methods.

- **Churn by Internet Service**
  A stacked bar chart was created to evaluate churn among internet service categories.

- **Churn Heatmap**
  A matrix heatmap was used to analyze churn patterns based on contract type and internet service combinations.

The dashboard enables users to explore customer churn trends and identify high-risk customer segments efficiently.

11. KEY FINDINGS

The analysis produced several important findings:

1. Month-to-month contract customers exhibit the highest churn levels.
2. Fiber Optic customers show higher churn compared to DSL customers.
3. Customers using Electronic Check payment methods contribute significantly to churn.
4. Customers with shorter tenure periods are more likely to leave.
5. Long-term contract customers demonstrate stronger retention and loyalty.

These findings highlight key areas where organizations can focus their retention efforts.

12. BUSINESS RECOMMENDATIONS

Based on the findings, the following recommendations are proposed:

- Encourage customers to adopt long-term contracts through incentives and loyalty programs.
- Improve customer experience for Fiber Optic users.
- Strengthen onboarding and retention programs for new customers.
- Monitor high-risk payment segments and develop targeted engagement strategies.
- Use dashboard insights to implement proactive customer retention initiatives.

13. DASHBOARD OVERVIEW

The final Power BI dashboard includes:

- Total Customers KPI Card
- Churned Customers KPI Card
- Churn Rate KPI Card
- Gender Distribution Analysis
- Partner Status Analysis
- Dependents Status Analysis
- Customer Tenure Histogram
- Churn by Contract Type
- Churn by Payment Method
- Churn by Internet Service
- Churn Heatmap

Dashboard Screenshot

14. TOOLS AND TECHNOLOGIES USED

- Microsoft Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Visualization Techniques
- Business Intelligence Reporting

15. CONCLUSION

The Customer Churn Analysis and Business Insights Using Power BI project successfully transformed raw customer data into meaningful business intelligence. Through data preparation, KPI development, demographic analysis, tenure analysis, and churn analysis, important customer retention patterns were identified.

The interactive dashboard provides stakeholders with a centralized platform for monitoring churn performance and understanding customer behavior. The insights and recommendations generated from this project can support strategic decision-making and help organizations reduce customer attrition while improving long-term customer retention.